

Lab Report No 1

Discrete Fourier Transform

Digital Image Processing



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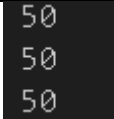
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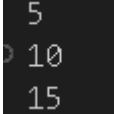
Lab Example No 1:

Solution:

Brief description (3-5 lines)
In this example we implement and run the code of Assigning single value to multiple variables in python.
The code
<pre>X = y = z = 50 print X print y print z</pre>
The results (Screenshot)


Lab Example No 2:

Solution:

Brief description (3-5 lines)
In this example we implement and run the code of Assigning multiple values to multiple variables in python.
The code
<pre>a,b,c=5,10,15 print a print b print c</pre>
The results (Screenshot)


Lab Example No 3:

Solution:

Brief description (3-5 lines)

In this example we implement and run the code of Python Data Types in python.

The code

```
a=10
b="Hi python"
c=10.5
print(type(a))
print(type(b))
print(type(c))
```

The results (Screenshot)

```
<class 'int'>
<class 'str'>
<class 'float'>
```

Lab Task No 2:

Solution:

Brief description (3-5 lines)

Write a Python code to solve the quadratic equation. (hint: import cmath library and use cmath.sqrt for this question)

The code

```
import cmath
def solve_quadratic(a, b, c):
    discriminant = (b ** 2) - (4 * a * c)
    sol1 = (-b + cmath.sqrt(discriminant)) / (2 * a)
    sol2 = (-b - cmath.sqrt(discriminant)) / (2 * a)
    return sol1, sol2
a = int(input("Enter coefficient a: "))
b = float(input("Enter coefficient b: "))
c = float(input("Enter coefficient c: "))
solution1, solution2 = solve_quadratic(a, b, c)
print(f"The solutions are {solution1} and {solution2}")
```

The results (Screenshot)

```
Enter coefficient a: 5
Enter coefficient b: 7
Enter coefficient c: 9
The solutions are (-0.7+1.1445523142259597j) and (-0.7-1.1445523142259597j)
```

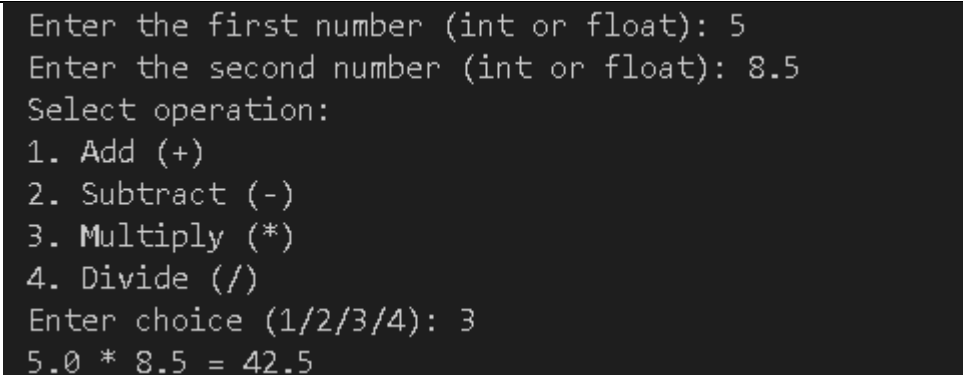
Lab Task No 3:

Solution:

Brief description (3-5 lines)
Write a Python code that implements basic calculator functioning (+, -, *, /) for user provided data of below mentioned data types. a. int b. float
The code
<pre>def add(x, y): return x + y def subtract(x, y): return x - y def multiply(x, y): return x * y def divide(x, y): if y == 0: return "Error! Division by zero." return x / y def calculator(): try: num1 = float(input("Enter the first number (int or float): ")) num2 = float(input("Enter the second number (int or float): ")) print("Select operation:") print("1. Add (+)") print("2. Subtract (-)") print("3. Multiply (*)") print("4. Divide (/)") choice = input("Enter choice (1/2/3/4): ") if choice == '1': print(f'{num1} + {num2} = {add(num1, num2)}') elif choice == '2': print(f'{num1} - {num2} = {subtract(num1, num2)}') elif choice == '3':</pre>

```
        print(f'{num1} * {num2} = {multiply(num1, num2)}')
    elif choice == '4':
        print(f'{num1} / {num2} = {divide(num1, num2)}')
    else:
        print("Invalid input! Please choose a valid operation.")
except ValueError:
    print("Invalid input! Please enter a number.")
calculator()
```

The results (Screenshot)



```
Enter the first number (int or float): 5
Enter the second number (int or float): 8.5
Select operation:
1. Add (+)
2. Subtract (-)
3. Multiply (*)
4. Divide (/)
Enter choice (1/2/3/4): 3
5.0 * 8.5 = 42.5
```

Conclusion

In the conclusion one of the most simple and readable language among high level programming languages Python is used in a great deal. These basic concepts are variables, data types, control structures (loops and branches), functions, and error handling. Python is multi-paradigm, it allows you to write code in many ways from procedural to object-oriented to functional programming. Python is great language for both newbie and experienced developers with its wide libraries and huge community. Learning this beginner material paves the way for higher-level subject matter and uses.